audible
an amazon company

Power Query Editor · Excel · Dataset Standardization

Data Analysis & Cleaning Project

Building an Automated Data Transformation Pipeline

9 Tasks

Audible Case Study

What Matters



Creating Better Listening Experiences through Clean Data

Audible connects millions of listeners with audiobooks, podcasts, and other spoken-word content. Behind every recommendation, search result, and business insight is a foundation of well-organized, reliable data.
With a vast catalog of books, authors, genres, release dates, and customer ratings, maintaining clean and consistent data is essential to:

- Deliver accurate reporting and analytics
- Improve search accuracy
- Generate personalized recommendations
- Enhance the overall user experience

Clean data is the first step toward trustworthy insights and better business decisions.

PROJECT OVERVIEW

Audible Data Cleaning

Utilize Power Query Editor in Excel to clean and standardize an Audible dataset. Tasks include formatting columns, ensuring data consistency, and preparing the dataset for analysis.

1. **Standardize Title Casing**
   Apply consistent title casing to book names.

2. **Clean Author Data**
   Separate multiple authors into individual fields.

3. **Format Release Dates**
   Convert all dates to DD-MM-YYYY format.

4. **Convert Time Values**
   Change text-based time into Excel duration.

5. **Validate Prices**
   Ensure numeric values and identify invalid entries.

6. **Standardize Ratings**
   Convert text ratings into numeric values.

7. **Split Narrator Details**
   Separate multiple narrators into different columns.

8. **Create Release Info**
   Merge release date and language into one field.

9. **Format Currency**
   Display all prices with two decimal places.

Q1 Standardize Title Casing

Column: name

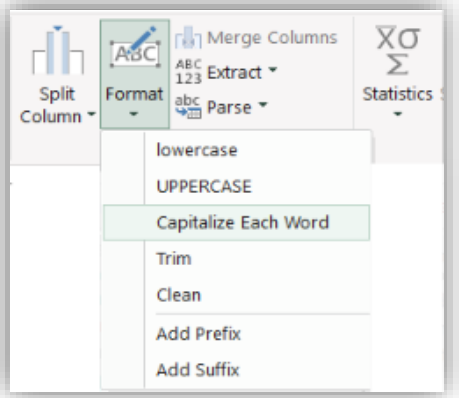
Task Question

Standardize the name column to ensure consistent title casing.

💡 Transform each entry in the name column so every word begins with a capital letter (Title Case), ensuring uniform presentation across all records.

Before

the PuffiN portal
thE GrunTs on The ruN
Utterly DaRk and tHe face OF the DeeP
The pecuLiar tale of tHe tentAcle BOY
a mouse cAlLed miiKA
nelL aNd the caVe bEar



After

The Puffin Portal
The Grunts on the Run
Utterly Dark and the Face of the Deep
The Peculiar Tale of the Tentacle Boy
A Mouse Called Miika
Nell and the Cave Bear

Q2

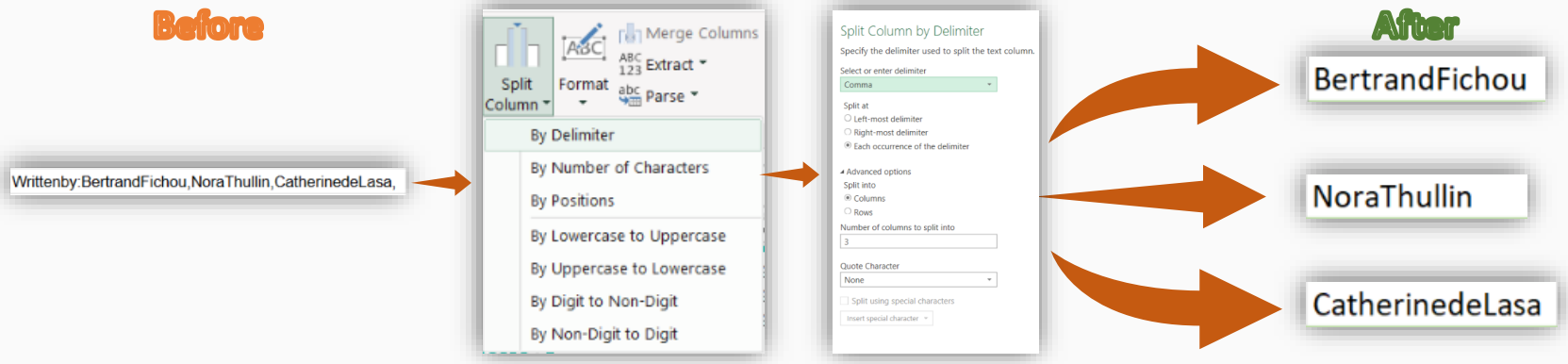
Separate Multiple Authors

Column: author

Task Question

Separate combined names in the author column if there are multiple authors.

💡 Identify records where multiple authors are listed together and split them into separate columns or rows for accurate attribution.



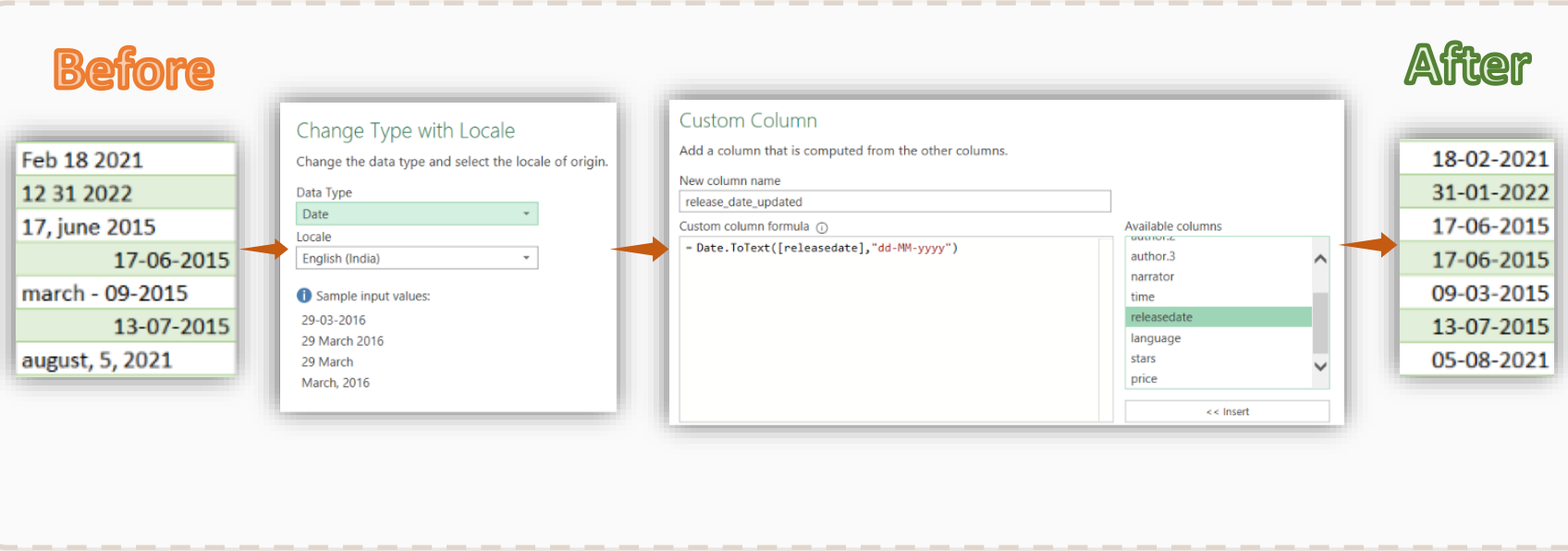
Q3 Consistent Date Format

Column: releasedate

Task Question

Ensure all entries in the releasedate column follow a consistent date format (DD-MM-YYYY).

Detect and convert any date entries that deviate from the DD-MM-YYYY format so all release dates are uniform across the dataset.



Q4 Convert Time to Duration

Column: time

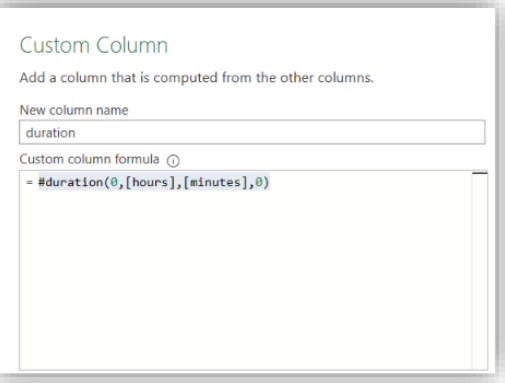
Task Question

Convert the time column from text format to a duration format that Excel recognizes.

Parse the text-based time values and transform them into Excel's native duration/time format, enabling time-based calculations.

Before

2 hrs and 20 mins
13 hrs and 8 mins
2 hrs and 3 mins
11 hrs and 16 mins
10 hrs
10 hrs and 35 mins
2 hrs and 23 mins



After

0.02:20:00
0.13:08:00
0.02:03:00
0.11:16:00
0.10:00:00
0.10:35:00
0.02:23:00

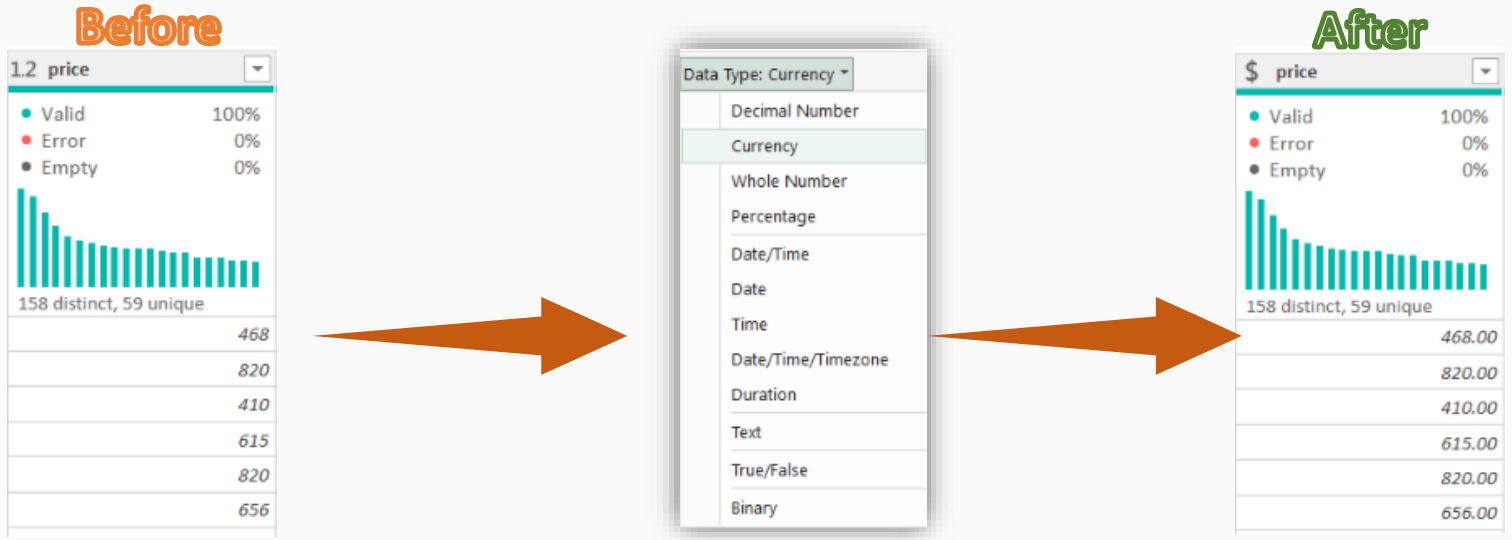
Q5 Numeric Price Validation

Column: price

Task Question

Ensure the price column is in a numeric format, and identify any non-numeric values.

Convert all price entries to numeric type, flag or isolate records containing text, currency symbols, or other non-numeric characters.



Q6

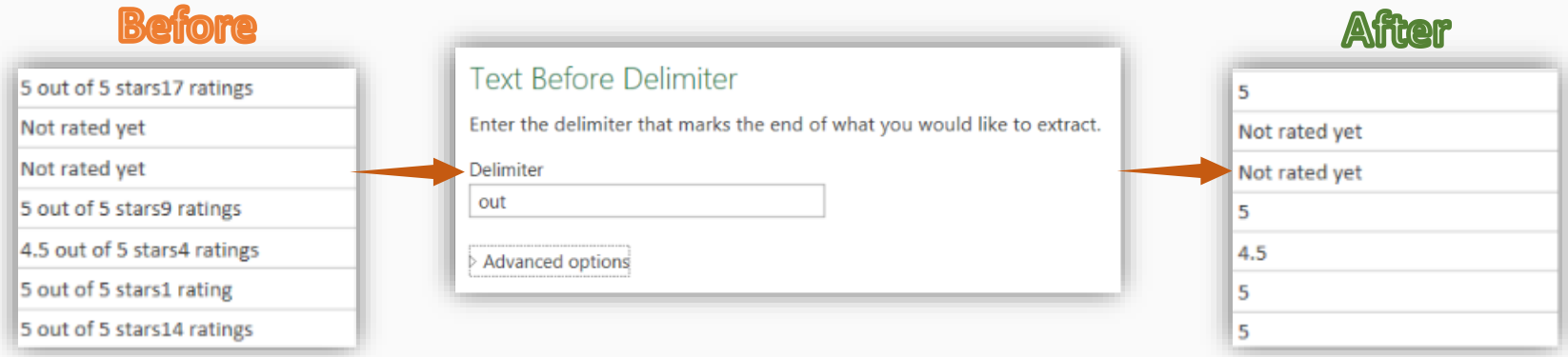
Convert Star Ratings to Numeric

Column: stars

Task Question

Convert text ratings in the stars column to numeric values.

💡 Replace any text-based star rating labels (e.g., 'Four and a half stars') with their corresponding numeric equivalents for quantitative analysis.



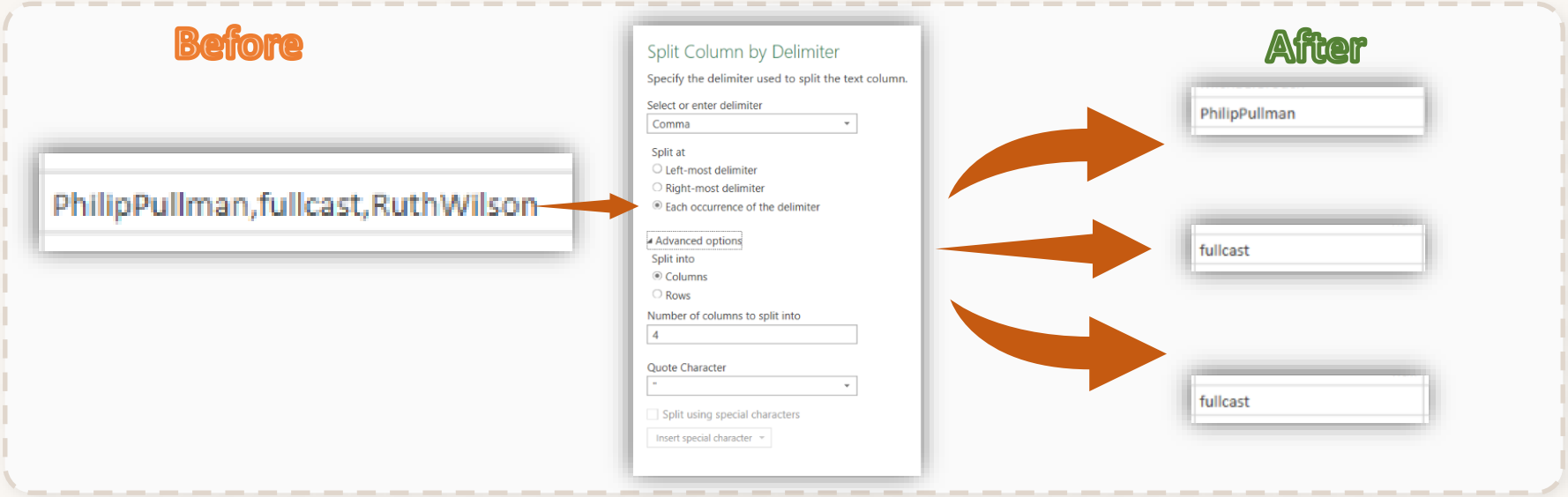
Q7 Split Multiple Narrators

Column: narratedby

Task Question

Split the narratedby column into multiple columns if multiple narrators are listed.

💡 For entries with more than one narrator, split the combined field into separate columns (Narrator 1, Narrator 2, etc.) to normalize the data.



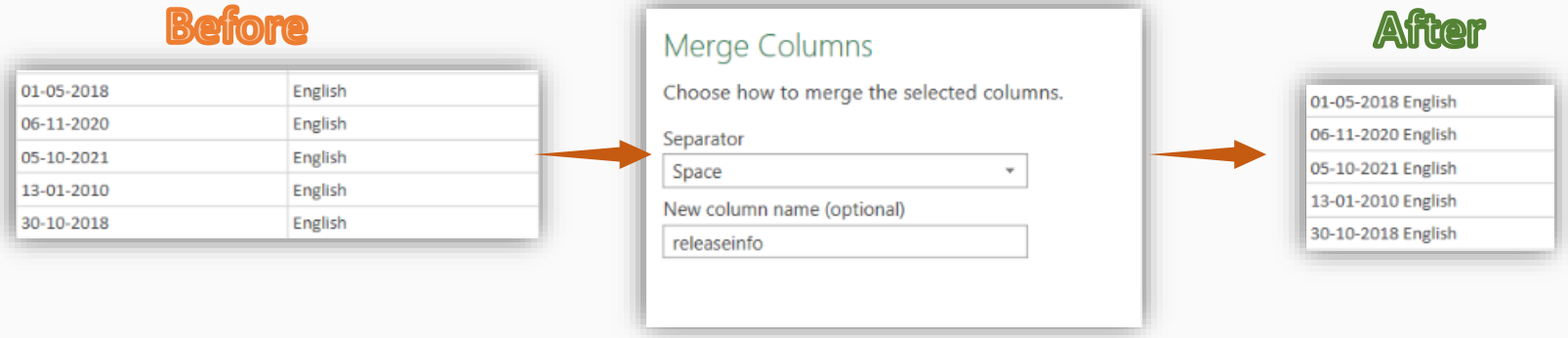
Q8 Merge releasedate & language

Column: releaseinfo

Task Question

Merge the releasedate and language columns into a single new column named releaseinfo with the format "DD-MM-YYYY, Language."

Create a new computed column releaseinfo by concatenating the formatted date and language fields with a comma separator.



Thank You

Vinay Mehra ✓

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